Blind image deconvolution with spatially adaptive total variation regularization

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A blind deconvolution algorithm with spatially adaptive total variation regularization is introduced. The spatial information in different image regions is incorporated into regularization by using the edge indicator called difference eigenvalue to distinguish edges from flat areas. The proposed algorithm can effectively reduce the noise in flat regions as well as preserve the edge and detailed information. Moreover, it becomes more robust with the change of the regularization parameter. Comparative results on simulated and real degraded images are reported. © 2012 Optical Society of America

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Observed images such as microscopy, medical, and astronomical imaging often suffer degradation by blurring. Blind deconvolution can be resorted to settle the problem. Assuming the imaging system to be linear and spatially shift-invariant, then the blurred noisy image $i(\mathbf{x})$ can be represented as:

$$i(\mathbf{x}) = h(\mathbf{x}) * o(\mathbf{x}) + n(\mathbf{x}), \tag{1}$$

where $\mathbf{x} = (x_1, x_2)$ denotes the spatial coordinates in the image, the symbol $*$ denotes two-dimensional convolution, $h(\mathbf{x})$ refers to the point spread function (PSF) of the degradation process, $o(\mathbf{x})$ represents the original image, and $n(\mathbf{x})$ is the noise. The goal of blind deconvolution is to seek the best estimates of $o(\mathbf{x})$ and $h(\mathbf{x})$ from the degraded image $i(\mathbf{x})$. In past decades, many blind deconvolution methods have been proposed (for a review see [1]).The Richardson–Lucy (RL) method proposed in [2,3] is the most well-known for image deconvolution. In [4], Fish *et al.* extended RL method to blind situation. Up to now, many modified or improved RL methods have been developed for image deconvolution in microscopy, medical, and astronomical imaging [5–8]. However, owing to the ill-posed nature of inverse problems, these algorithms lack stability and uniqueness, consequently, they require regularization. The total variation (TV) was initially introduced as a regularizer for image restoration in [9]. Since then it has been used extensively and with great success for inverse problems because of its ability to preserve edges. Nevertheless, TV model has certain shortcomings. First, it favors a piecewise constant solution, which produces “staircase effects” in flat regions of the image, and some spatial features, such as textures and small details, are often diminished with noise. Second, TV-based method is sensitive to the regularization parameter. For instance, a large regularization parameter can help reduce the noise in flat regions, but blur the edge and detailed information at the same time, which limits the application in practice. To overcome these shortcomings, an improved RL algorithm with spatially adaptive TV regularization (RLSATV) is introduced. The proposed RLSATV method can automatically balance the regularized strength between different spatial property regions in the image. Consequently, RLSATV can effectively reduce the noise in flat regions as well as preserve the edge and detailed information. Moreover, RLSATV becomes more robust with the change of the regularization parameter.

Considering the cases where the data is contaminated by Poisson noise, the intensity of each pixel $\mathbf{x}$ in the observed image is a random variable that follows an independent Poisson distribution. The likelihood distribution can be given by:

$$P(i|o,h) = \prod_{\mathbf{x}} \frac{[(h*o)(\mathbf{x})]^{i(\mathbf{x})} \exp(-(h*o)(\mathbf{x}))}{i(\mathbf{x})!}. \tag{2}$$

To estimate o and h, iterative deconvolution algorithms can be used. One option is to use RL algorithm. The nonregularized RL algorithm minimizes the functional, giving the maximum likelihood estimation:

$$J_1(o) = \sum_{\mathbf{x}} (-i(\mathbf{x}) \log[(h*o)(\mathbf{x})] + (h*o)(\mathbf{x})). \tag{3}$$

RL algorithm does not converge to the solution because the noise is amplified after a small number of iterations. In order to get a better convergence, RL algorithm with total variation regularization (RLTV) was suggested in [5,6], and the total functional to be minimized is then

$$J_2(o) = \sum_{x} (-i(\mathbf{x}) \log[(h*o)(\mathbf{x})] + (h*o)(\mathbf{x})) + \lambda \sum_{\mathbf{x}} |\nabla o(\mathbf{x})|. \tag{4}$$

In Eq. (4), λ is the regularization parameter, which plays a very important role. It controls the TV regularization strength. If λ is too small, the noise will not be well suppressed; inversely, if it is too large, the edge and detailed information in image will be blurred. It means that the regularization parameter λ is spatially dependent. Therefore, in this letter, a spatially weighted TV regularization model considering the spatial dependent property of the TV regularization is introduced. A critical problem is to select a good spatial information indicator, which can distinguish edges from flat areas in the image in the presence of noise. To do this, we use the spatial information extractor called difference eigenvalue from Tian [10] to indicate the spatial information. The difference eigenvalue is based on the Hessian matrix of the image [11]. We use a Gaussian filtered version of the Hessian matrix to improve the robustness to noise:

$$J_\rho = \begin{bmatrix} j_{11} & j_{12} \\ j_{12} & j_{22} \end{bmatrix} = \begin{bmatrix} o_{x_1x_1} * G_\rho & o_{x_1x_2} * G_\rho \\ o_{x_1x_2} * G_\rho & o_{x_2x_2} * G_\rho \end{bmatrix}, \quad (5)$$

where G_ρ denotes the Gaussian kernel with the parameter ρ (the size is 5×5 and $\rho = 0.8$ in this letter). The two eigenvalues of the J_ρ, denoted by λ_1 and λ_2, are given by

$$\lambda_{1,2} = \frac{1}{2}\left[(j_{11} + j_{22}) \pm \sqrt{(j_{11} - j_{22})^2 + 4j_{12}^2}\right]. \quad (6)$$

Let λ_1 denote the larger eigenvalue and λ_2 denote the other eigenvalue. Here λ_1 and λ_2 correspond to the maximum and minimum local variation at a pixel, respectively. The modified difference eigenvalue edge indicator $D(\mathbf{x})$ is defined as:

$$D(\mathbf{x}) = (\lambda_1 - \lambda_2)\lambda_1 w(o(x_1, x_2)), \quad (7)$$

where $w(o(x_1, x_2))$ is a weighting factor which is used to achieve balance between detail enhancement and noise suppression. Its value is estimated from the gray level variance and defined by

$$w(o(x_1, x_2)) = \frac{\sigma(x_1, x_2) - \min(\sigma)}{\max(\sigma) - \min(\sigma)}, \quad (8)$$

where $\min(\sigma)$ and $\max(\sigma)$ are minimum and maximum gray level variances of image o, respectively. For a given pixel $o(x_1, x_2)$, the gray level variance is calculated from its 3×3 neighborhood

$$\sigma(x_1, x_2) = \frac{1}{9}\sum_{i=-1}^{1}\sum_{j=-1}^{1}[o(x_1 + i, x_2 + j) - o(x_1, x_2)]. \quad (9)$$

Combining the edge indicator $D(\mathbf{x})$ into TV model, we define a novel SATV model with spatially adaptive property:

$$\mathrm{SATV} = \sum_{\mathbf{x}} \frac{1}{1 + \beta D(\mathbf{x})} |\nabla o(\mathbf{x})|, \quad (10)$$

here β is a constant.

Substituting the TV term in Eq. (4) with SATV in Eq. (10), we introduce the cost functional:

$$L(o, h) = \sum_{\mathbf{x}} \left\{ -i(\mathbf{x}) \log(h * o)(\mathbf{x}) + (h * o)(\mathbf{x}) + \frac{\lambda}{1 + \beta D(\mathbf{x})} |\nabla o(\mathbf{x})| \right\}. \quad (11)$$

Minimizing Eq. (11) by the expectation-maximization algorithm, we obtain the novel RLSATV iterative algorithm:

$$h_t^{(n+1)}(\mathbf{x}) = h^{(n)}(\mathbf{x}) \left\{ o^{(n)}(-\mathbf{x}) * \left[\frac{i(\mathbf{x})}{h^{(n)}(\mathbf{x}) * o^{(n)}(\mathbf{x})} \right] \right\},$$
$$h^{(n+1)}(\mathbf{x}) = \frac{h^{(n+1)}(\mathbf{x})}{\sum_{\mathbf{x}} h_t^{(n+1)}(\mathbf{x})}, \quad (12)$$

$$o^{(n+1)}(\mathbf{x}) = o^{(n)}(\mathbf{x}) \left\{ h^{(n+1)}(\mathbf{x}) * \left[\frac{i(\mathbf{x})}{h^{(n+1)}(\mathbf{x}) * o^{(n)}(\mathbf{x})} \right] \right\} \times \frac{1}{1 - \frac{\lambda}{1+\beta D(\mathbf{x})} \mathrm{div}\left(\frac{\nabla o^{(n)}(\mathbf{x})}{|\nabla o^{(n)}(\mathbf{x})|} \right)}. \quad (13)$$

Summarizing, two good features of RLSATV can be drawn: (1) for flat area pixels because $D(\mathbf{x})$ is close to zero, $1/(1 + \beta D(\mathbf{x}))$ is close to 1, which means a large TV regularization strength is enforced to these pixels, and then the noise will be well suppressed; (2) for edge and detailed pixels because $D(\mathbf{x})$ is very large, $1/(1 + \beta D(\mathbf{x}))$ is small and almost close to zero, and this weakens the TV regularization strength enforced to these pixels, so the edge and texture will be well preserved. That is, RLSATV has the ability to adaptively adjust the regularization strength according to the spatial property of the image.

To test the performance of RLSATV, we executed simulations with three test images (because of page limitation, test images are omitted here) under the Poisson noise process. The images were degraded by the size 15×15 Gaussian blur kernel with the standard deviation of 2. Then the blurred image was contaminated by Poisson noise. For comparison, RL and RLTV algorithms were also tested. To measure the improvement in the restored image quality, the normalized mean square error (NMSE) $\|o - \hat{o}\|^2 / \|o\|^2$ is used, where o and $\hat{o}$ are the original and the restored image, respectively.

Figure 1 shows a simulation example of restoration by RLTV and RLSATV. We chose $\lambda = 0.0155$, $\beta = 2$, and 300 iterations. Comparing Fig. 1(c) with Fig. 1(d), it can be found that RLSATV obviously reduces the artifacts caused by the noise as well as better preserves the edges. The effects of the regularization parameter were also investigated. For RLTV and RLSATV methods, the NMSE of the restored image versus varying regularization parameter λ on the simulated degraded image of Fig. 1(b) are plotted in Fig. 2. RLSATV method appears more robust with the change of the regularization parameter at a

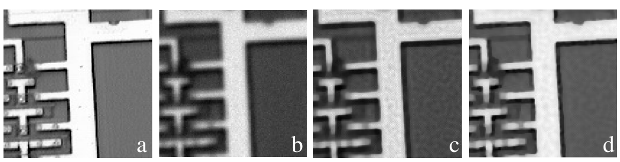

Fig. 1. Restoration of a simulated degraded image. (a) Original circuit image; (b) blurred and noisy image; (c) image restored by RLTV; and (d) image restored by RLSATV.

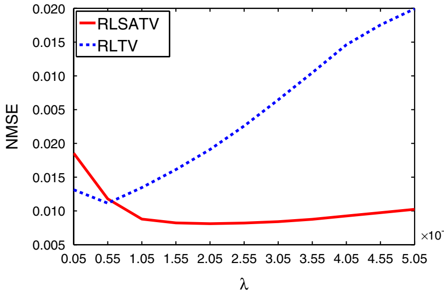

Fig. 2. (Color online) NMSE versus regularization parameter λ of RLTV and RLSATV for the circuit image.

Table 1. NMSE of Degraded Image and the Best Restored Image (with the Lowest NMSE) by Different Algorithms

		Image Restoration by		
Image	Degraded Image	RL	RLTV	RLSATV
satellite	0.0615	0.0395	0.0404	0.0383
circuit	0.0203	0.0116	0.0085	0.0072
liver	0.0467	0.0218	0.0185	0.0184

wide range from 0.0005 to 0.0505 at intervals of 0.0005. However, the change of the regularization parameter λ has a very large effect on the performance of RLTV. In particular, the NMSE value increases abruptly when the regularization parameter becomes large. To show the effect of the parameter β on the RLSATV method, we tested the NMSE values when β was selected to be 0.1, 0.2, 0.5, 1, 2, 5, and 10, respectively. For each test image, the NMSE values change little with the variation of the parameter β and achieve little lower values when β ranges from 1 to 5. Therefore, for all the experiments in this letter, the parameter β was set to be 2.

Moreover, the NMSE of degraded images and the best restored images by the three methods are compared in Table 1. The best restored image is selected to be the one with the lowest NMSE when the regularization parameter changes. It can be seen that the RLSATV method has achieved the smallest NMSE among the three methods, and NMSE of the RLSATV method is considerably smaller than that of the degraded image for every image. The NMSE versus the number of iteration of the three methods on the "circuit" image are plotted in Fig. 3, where the convergence of RLSATV is also highlighted.

An example of a real degraded image is shown in Fig. 4. As for the parameters, we use $\lambda = 0.02$, starting with 25×25 Gaussian kernel with the standard deviation of 1.5 for the initial estimate of PSF and 100 iterations. Figure 4(b) is the restored result by RLTV, apparent piecewise constant effect can be seen, for instance on the ring. Figure 4(c) is the restored result by RLSATV. Obviously, more fine details are restored. The curve of relative error for the estimations of the successive iteration (RESE) $\|o^{(n+1)} - o^{(n)}\| / \|o^{(n)}\|$ versus iteration number is shown in Fig. 4(d). Owing to the spatially adaptive regularization, the proposed RLSATV achieves a better convergence property.

In conclusion, an improved blind deconvolution algorithm with spatially adaptive TV regularization (RLSATV) is proposed. The proposed method can automatically

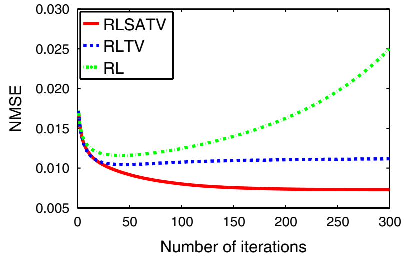

Fig. 3. (Color online) NMSE versus the iteration number of the three methods for the circuit image with $\lambda = 0.0155$.

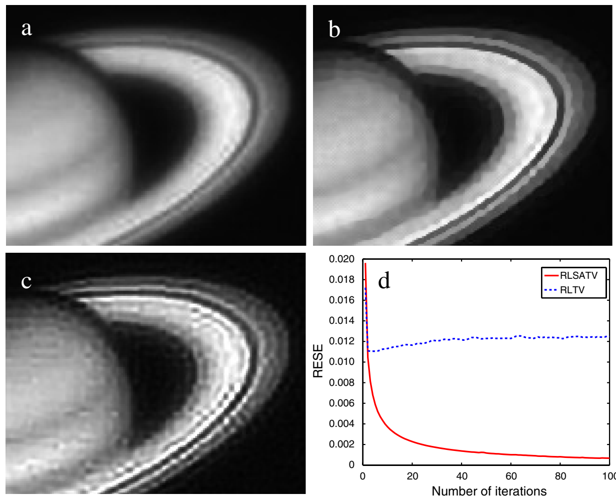

Fig. 4. (Color online) Restoration of a real degraded image. (a) Real degraded image, (b) image restored by RLTV, (c) image restored by RLSATV, and (d) RESE versus iteration number.

balance the regularized strength between different spatial property regions in the image. Comparative results on simulated and real images show that RLSATV can effectively reduce the noise in flat regions as well as preserve the edge and detailed information. Moreover, it becomes more robust with the regularization parameter and apt to converge, which make the method more preferable in practical applications.

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